

From Despair to Hope: Breakthrough in Spinal Metastasis Surgery Using Advanced Classification System

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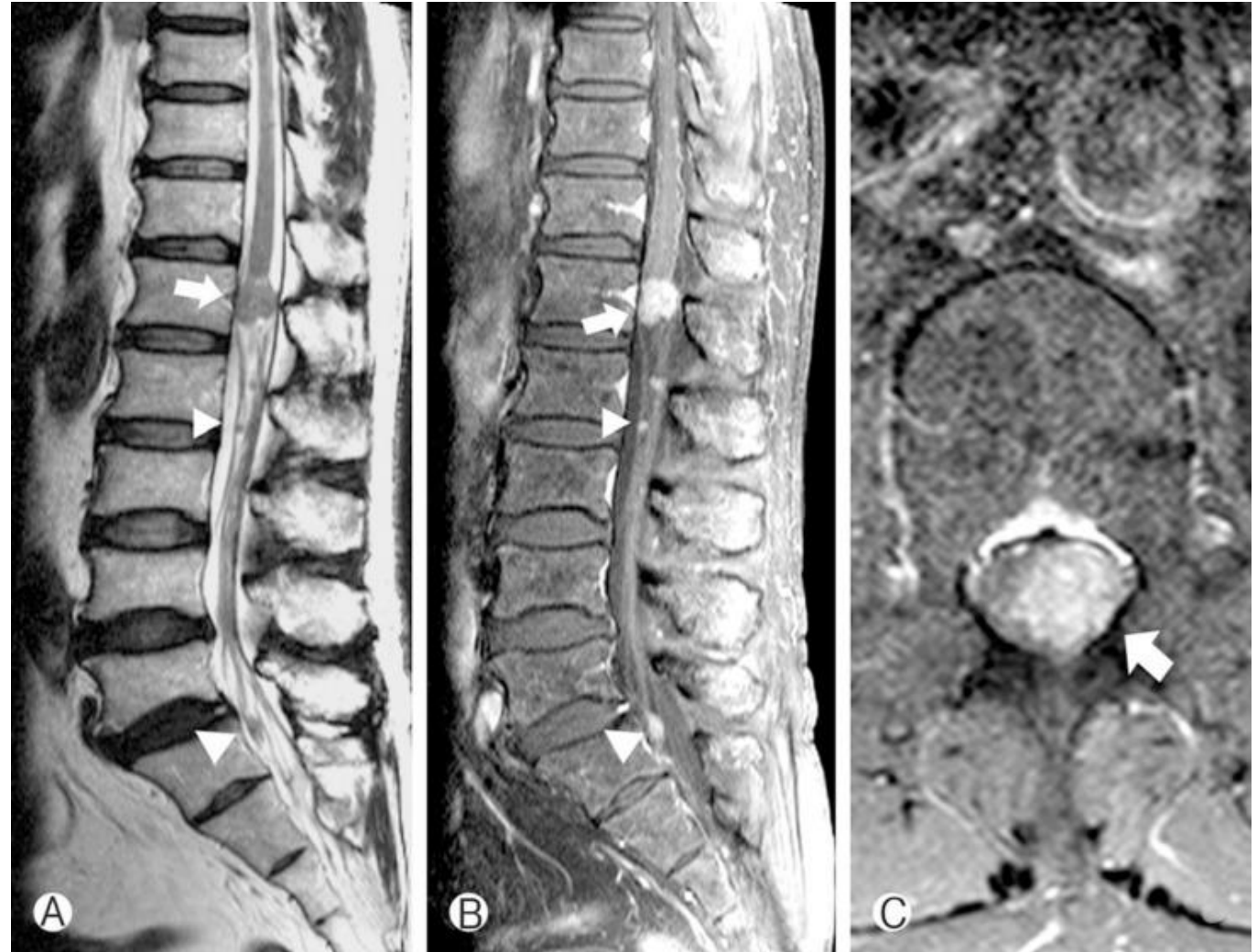
Azharuddin, MD, PhD

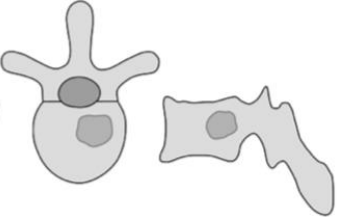
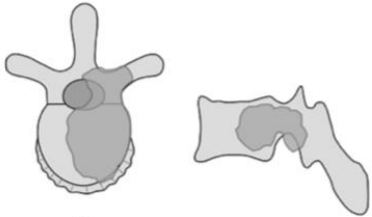
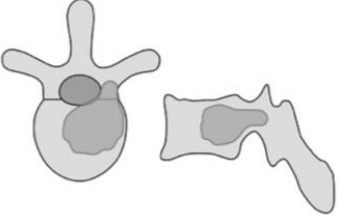
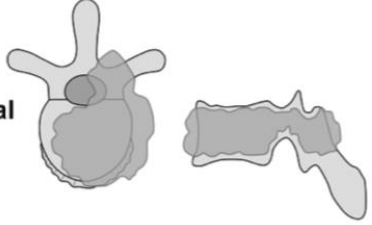
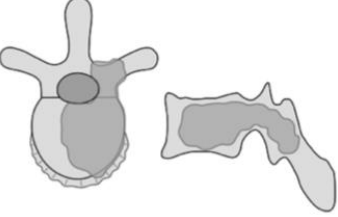
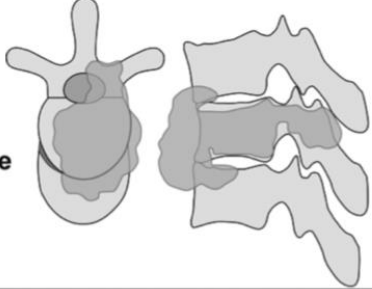
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**Orthopaedi & Traumatologi
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- Metastatic spinal tumors is rising due to the extended survival periods of cancer patients
- Spinal metastasis presents significant **challenges**
- Patients with neurological deficits and an expected lifespan of three months or more, surgery is often recommended
- Classification system are essential tools for assessing disease extent and guiding surgical decisions



Intra-compartmental	Extra-compartmental	Multiple
Type 1 vertebral body 	Type 4 epidural extension 	Type 7 
Type 2 pedicle extension 	Type 5 paravertebral extension 	
Type 3 body-lamina extension 	Type 6 2-3 vertebrae 	

Tomita Classification

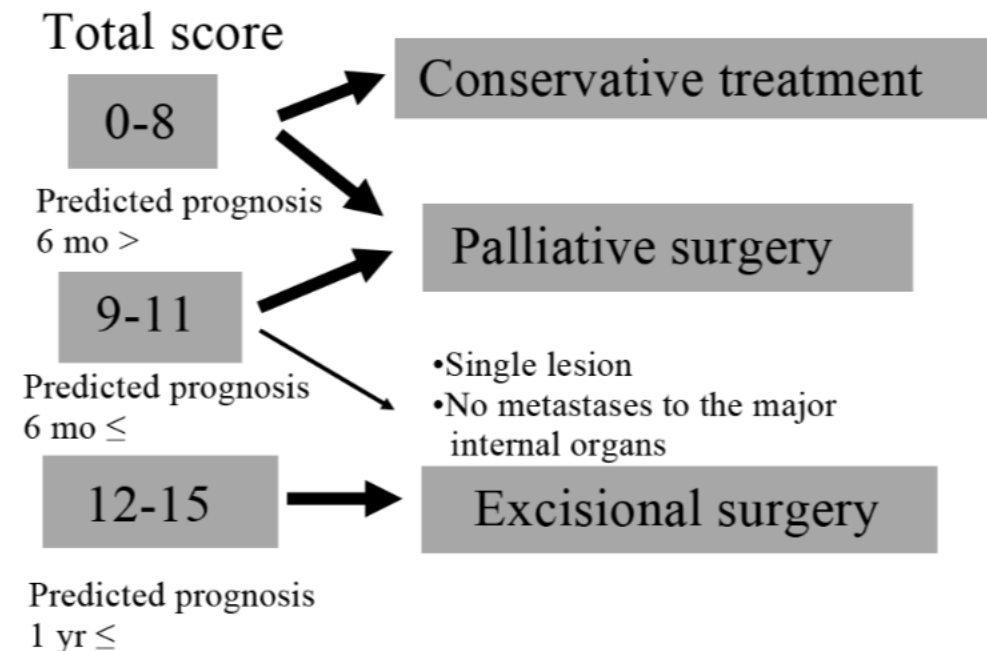
Harrington Classification

Classification	Neurologic symptom and degree of bone destruction
Class 1	Minimal neurology
Class 2	Involvement of bone without collapse or instability and minimal neurology
Class 3	Major neurologic impairment without spinal instability
Class 4	Vertebral collapse and instability, without major neurologic impairment
Class 5	Vertebral collapse and instability with major neurologic impairment
Treatment principle	
Class 1, 2	Chemotherapy, hormonal therapy, radiotherapy
Class 3	Radiotherapy with corticosteroid treatment
Class 4, 5	Surgery

Characteristic	Score
General condition (performance status, %)	
Poor (10–40)	0
Moderate (50–70)	1
Good (80–100)	2
No. of extraspinal bone metastases foci	
3 or more	0
2	1
1	2
Metastases to the major internal organs	
Unremovable	0
Removable	1
No metastases	2
Primary site of the cancer	
Lung, osteosarcoma, stomach, bladder, esophagus, pancreas	0
Liver, gallbladder, unidentified	1
Others	2
Kidney, uterus	3
Rectum	4
Thyroid, breast, prostate, carcinoid tumor	5
Palsy	
Complete (Frankel A, B)	0
Incomplete (Frankel C, D)	1
Non (Frankel E)	2

Criteria of predicted prognosis: total score (TS) 0–8, less than 6 months; TS 9–11, 6 months or more; TS 12–15, 1 year or more.

Predicted Prognosis Score



Case

- 51 year old male
- History of Thyroid Cancer
- Burst Fracture of Thoracal XI, Frankel C
- Multiple metastasis on Thoracal VIII and Lumbar 3

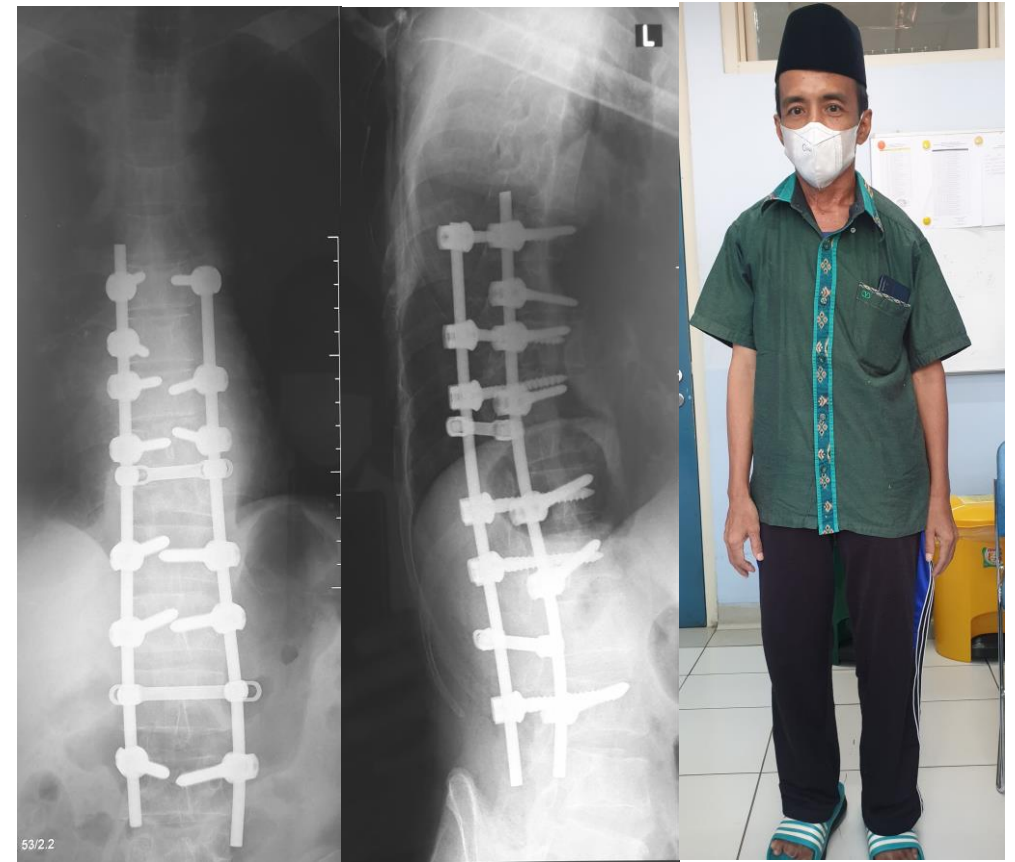
Classification

- Tomita Type 7
- Harrington Class 5
- Predicted Prognosis Score 10



Performed

- Posterior Spinal Fusion
- Frankel C to Frankel D on six months follow up



Summary

- This case underscores the value of the Tomita classification and Harrington classification in planning and executing surgical interventions
 - Whether the patient should receive surgery or only conservative therapy can be determined and discussed during informed consent or preoperatively before the procedure.
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Thank You
